HIREVERSE

Job and Internship Portal

Project Documentation

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Batch: Full Stack Development with Java

ABSTRACT

HireVerse is a professional full-stack job and internship portal designed to bridge the gap between candidates and recruiters. Utilizing a secure Spring Boot backend and a dynamic React frontend, the platform streamlines every stage of the recruitment lifecycle. Its centerpiece is an automated ATS scoring system that ensures data-driven, efficient screening and transparent application tracking, providing a modern solution for today's competitive job market.

1. PROBLEM STATEMENT

Finding a job or a good candidate is very hard today because many portals are slow and disorganized. Recruiters often get hundreds of resumes for one job and cannot check them all by hand. At the same time, candidates apply for jobs but never get any updates about whether they are selected or not. Manually matching skills between a person and a job takes too much time, and using many different tools makes it difficult for a company to manage everything in one place. This creates a big gap that HireVerse aims to fix.

2. PROPOSED SOLUTION - ARCHITECTURE

HireVerse provides a simple solution by bringing everything together in one website. It uses a modern architecture where the frontend and backend work together to keep data secure and the website fast.

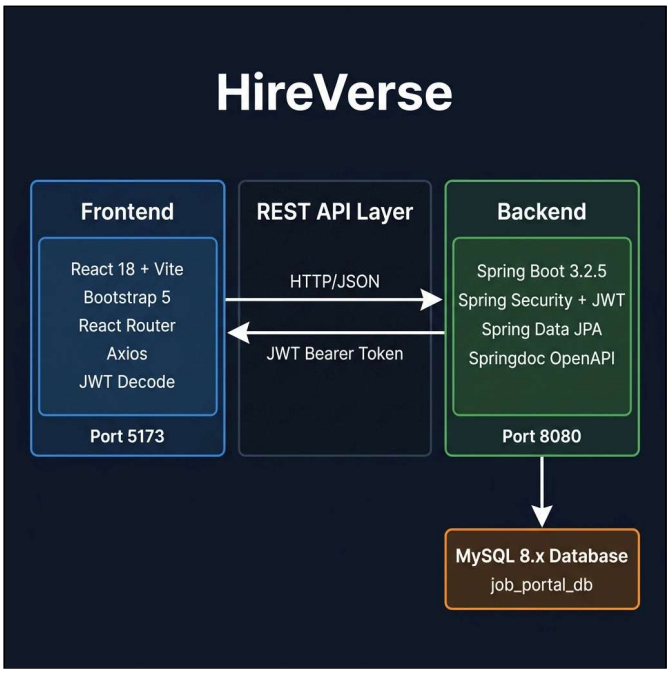

Figure 1: High-Level System Architecture (Frontend - Backend - Database)

3. CORE FEATURES & FUNCTIONALITIES

The HireVerse platform is designed with role-specific functionalities to ensure a smooth journey for all types of users. The following table summarizes the key features provided in each module:

Module	Key Features & Functionalities
Candidate	Registration, Secure Login, Profile builder (Bio/Skills/Education), Advanced job discovery with filters, Stateless application tracking, Instant status notifications.
Recruiter	Job lifecycle management (Post/Edit/Close), Applicant dashboard, Automatic sorting by ATS score, Multi-stage hiring pipeline (Shortlist, Interview, Selection).
Admin	Platform-wide statistics (Users, Jobs, Apps), User lifecycle moderation (Ban/Unban), Content management (Job categories), Platform health monitoring.

4. DATABASE DESIGN & SCHEMA

The database architecture utilizes normalized relational principles to ensure data consistency and integrity across the platform's multi-role ecosystem.

Entity	Primary Table	Key Attributes / Relationships
User	users	ID, Name, Email (Unique), Password (Hashed), Role, isActive status.
Profile	candidate_profiles recruiter_profiles	ID, UserID (FK), Bio, Skills, Experience / Company, Designation. Linked 1:1 with User.
Job	job_listings	ID, Title, Description, RecruiterID (FK), CategoryID (FK), Status, Deadline.
Application	applications	ID, CandidateID (FK), JobID (FK), Status, ATS Score, Cover Letter.

5. TECHNOLOGY STACK

Layer	Technology Used	Project Role
Backend	Spring Boot 3 / Java 17	RESTful API, Security, Business Logic.
Frontend	React 18 / Vite	Single Page Application (SPA), Component UI.
Database	MySQL 8.x / JPA	Persistent storage and ORM mapping.
Security	Spring Security / JWT	Stateless role-based access control.

6. UNIQUE POINTS: AUTOMATED ATS SCORING

The integrated ATS engine calculates a compatibility percentage based on the match between a candidate's profile skills and a job's requirements instantly upon application. This helps recruiters prioritize the most suitable candidates without manual intervention.

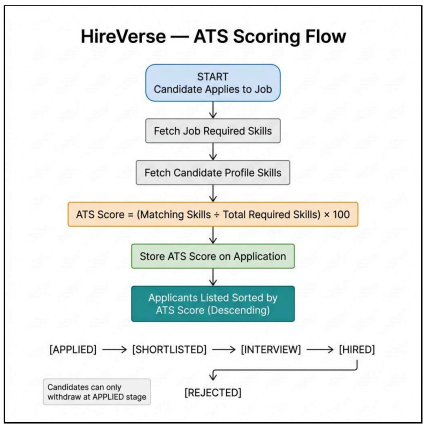

Figure 2: Automated ATS Scoring & Recruitment Pipeline Logic

7. SCREENSHOTS

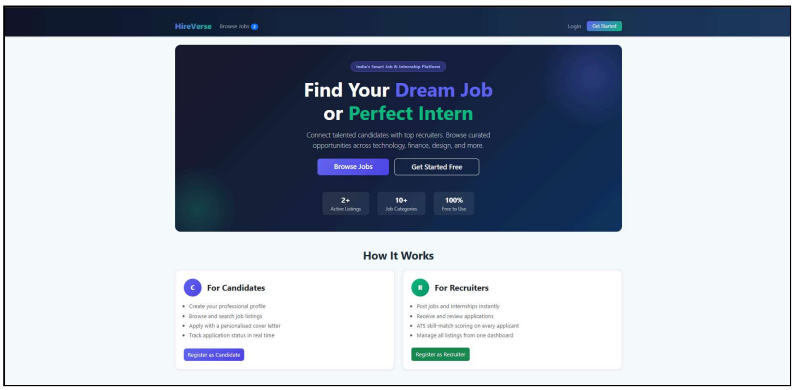

Figure 3: HireVerse Home Page

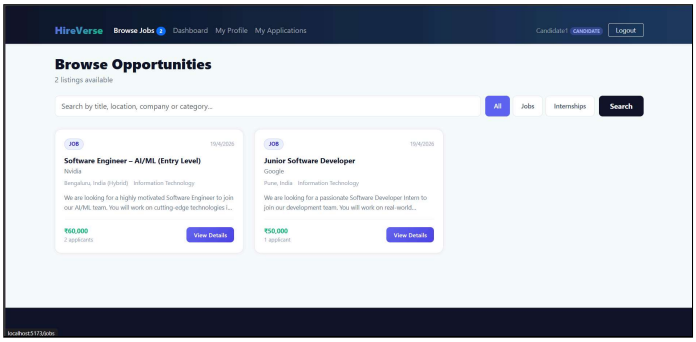

Figure 4: Job Search and Filtering Interface

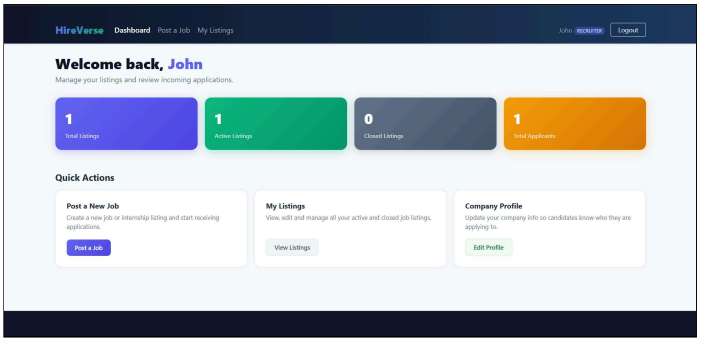

Figure 5: Recruiter Management Dashboard

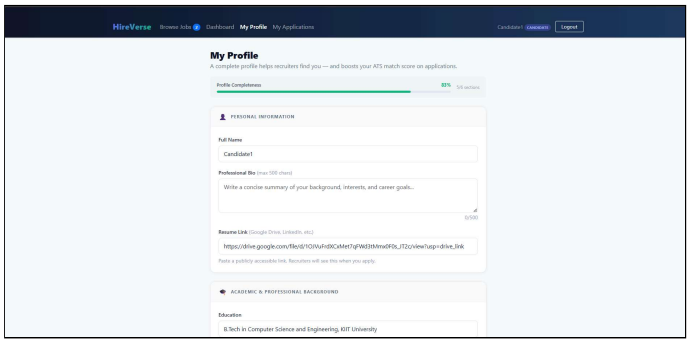

Figure 6: Candidate Profile and Skill Management

8. FUTURE IMPROVEMENTS

- **AI-Based Resume Parsing:** Integration of NLP modules to automatically extract candidate data from PDF resumes, reducing manual profile entry time by up to 80%.
- **Real-time Messaging System:** Implementation of WebSockets to provide a direct chat interface between recruiters and shortlisted candidates for faster interview scheduling.
- **Advanced Hiring Analytics:** Development of a business intelligence dashboard for recruiters to track hiring velocity and source effectiveness.

9. CONCLUSION

HireVerse successfully delivers a modern recruitment solution, combining technical robustness with an optimized user experience. By utilizing the Spring Boot and React stack, the project demonstrates a complete, end-to-end integration of secure authentication, data persistence, and dynamic UI. The automated ATS scoring engine provides a significant value proposition, making the recruitment process more efficient, transparent, and data-driven for all stakeholders involved.